

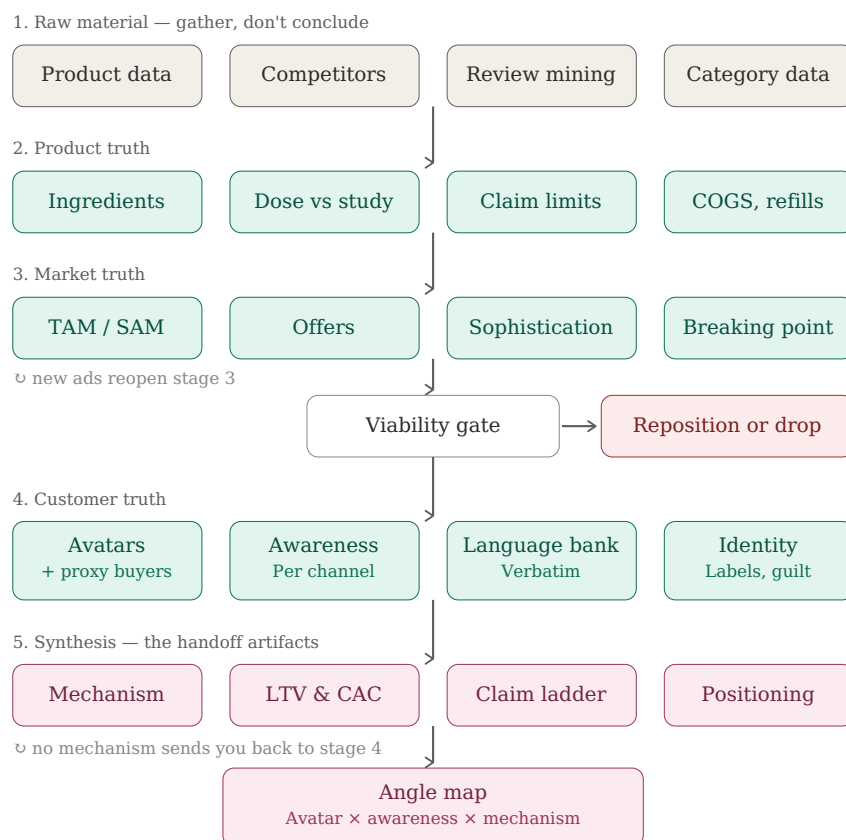
The research compartment

A reusable framework for supplement subscription businesses

Stage one of the marketing production cycle · product-agnostic · version 2

Research is usually treated as a checklist of six or seven things to find out. It isn't. It is a dependency graph, and several of its edges only run one way — you cannot assess market sophistication before you have mapped the competitive landscape, and you cannot construct a mechanism before you know the sophistication stage you are writing against. Run the items in the wrong order and you get answers that look complete and are quietly wrong.

This document lays out the compartment as a flow: five stages, twenty-one nodes, one gate, and one output artifact. Each node below carries a purpose, the artifact it produces, and the condition under which it is finished. It is written to be executed either by a person or by an agent, which is why the done-criteria are stated in terms of observable output rather than effort.



The research compartment. Gray nodes are gathered inputs, teal nodes are derived findings, pink nodes are synthesis outputs.

Three edges that only run one way

Competitor landscape → sophistication. Sophistication is not a property you assess by intuition. It is a description of what the market has already been sold, which means it is read off the competitive landscape or it is guessed.

Sophistication + product truth → mechanism. The mechanism's job is to out-flank the claims already running. Building one before you know what stage you are at produces a mechanism that is either unnecessary or already used.

Claim limits → everything downstream. What is legally sayable is a ceiling, not a filter applied at review. Establish it once, in research, and nothing gets written that is not on the ladder.

Gathering only. The single most common failure in this compartment is concluding while collecting — reading three competitor ads and deciding the market is stage four. Keeping stage one free of judgements means every conclusion downstream has a traceable source, which is what makes the work auditable when someone reviews it.

Product data

The only node in this stage that does not come from the web. It comes from the supplier spec sheet, the certificate of analysis, and the physical label. That changes its done-criteria: the other three are done when they saturate, this one is done when it is complete. A missing field here is a hole, not a diminishing return.

Capture the formula panel exactly as printed — every ingredient in label order with per-serving amount and unit; the specific form of each active, since magnesium glycinate and magnesium oxide are different products commercially even where they are the same element; standardisation and extract ratios; branded ingredients such as KSM-66 or Creapure, which matter disproportionately because they carry their own literature and their own permitted claim language; proprietary blends flagged as such with total blend weight; excipients and capsule material, which generate more objections than the actives do; allergens; serving size, servings per container, and directions.

Then the physical and sensory facts — pill size, taste, smell, mixability, bottle format, shelf life. Pill size and taste are among the largest churn drivers in supplements and are recorded nowhere else in the flow. Then supply chain: manufacturer, country, facility certifications, whether a COA can actually be produced on demand, MOQ, lead time, unit cost, landed cost. Then commercial state: current SKUs, prices, subscription discount, bundles, and existing review volume if the product is live.

Output: a complete structured product record with source attribution per field.

Done when: every field is populated or explicitly marked unavailable. Proprietary blends are marked as gaps rather than estimated.

Guard: the supplier's marketing one-pager is not the dossier. It rounds doses and describes blends by name. Where it disagrees with the COA, the COA wins.

Competitors

Two sets, kept separate. The direct set sells the same formulation category to the same buyer. The adjacent set solves the same problem by a different route — a different supplement, a device, a prescription, a behaviour change. The adjacent set is where repositioning options come from later, and it is the set people skip.

For each competitor, pull the live creative from the platform ad libraries, the funnel type and its entry point, the landing page structure, the price ladder, the subscription discount, guarantees, and the upsell path. Store the artifacts themselves — ad copy, screenshots, page text — not summaries of them. Sophistication assessment and mechanism construction both read from this evidence directly, and a summary written now will have discarded exactly the detail they need.

Record first-seen dates alongside the creative. Longevity is the only performance signal available from outside a competitor's ad account — an ad still running after four months is winning, because nobody funds a loser that long. Capture the hook construction and the production style too: whether the opening measures the size or the speed of the claim, calls out an identity, challenges a belief, or leads with authority; whether the execution is studio or real-world, professional or user-generated, benefit-led or problem-led. These are observations about what exists, so they belong in gathering. What to conclude from them is the analyst's job in the next compartment, not yours.

Output: a competitor file per brand with archived creative, funnel artifacts, and first-seen dates.

Done when: the direct set is exhaustive for meaningful spend, and new ad library pulls stop surfacing new angles.

Review mining

Your own reviews if the product is live, and competitor reviews always. Amazon, Trustpilot, the brands' own sites, Reddit threads, TikTok and YouTube comments. Supplements are unusually well served here because buyers report both efficacy and experience in the same breath.

Capture verbatim text with star rating and date attached. Tag along three separate axes rather than one: why they bought, why they stayed, and why they quit. These are different populations and collapsing them into a single sentiment score destroys the distinction that retention work later depends on. Low-star reviews on competitor products are the highest-value material in the whole stage — they are your objection list and your differentiation list simultaneously.

Output: a tagged corpus of verbatim reviews, three-axis coded.

Done when: new reviews stop producing new themes. Typically several hundred, weighted toward one and two star.

Category data

Search volume for the category and for the head symptoms and ingredients, the trend line over several years, the seasonality curve, and whatever revenue estimates are obtainable for the category and for individual sellers. Retail versus direct split matters for a subscription business, because a category dominated by retail has different acquisition economics than one that grew online.

Output: a data table with sources and dates, no interpretation attached.

Done when: the market-sizing node has the inputs it needs to compute rather than estimate.

Turning the dossier into assessments. Everything here is about the product in isolation — no market comparison yet.

Ingredients

Mechanism of action per active, written in plain language a non-scientist can follow. Not the marketing description — the actual pathway. Which receptor, which enzyme, which deficiency corrected, over what timescale, with what expected magnitude of effect.

Two things to record that people leave out. First, time to effect: an ingredient that works in forty minutes and one that works in eight weeks demand completely different offers and completely different retention strategies. Second, which actives carry the story. Most formulas have one or two ingredients doing the narrative work and five doing supporting work, and knowing which is which prevents the copywriter from splitting attention across all seven.

Output: a mechanism note per active, with time-to-effect and a story-weight designation.

Done when: each active has a pathway explanation that survives a scientifically literate reader.

Dose vs study

For each active, your per-serving dose set against the dose used in the trials that established the effect, expressed as a ratio. This produces the proof strength for every claim you might later want to make, and it is the difference between a claim you can substantiate and one you can only assert.

Underdosing is common and worth being unsentimental about — an ingredient at fifteen percent of the studied dose is a label decoration, and treating it as a claim source is how brands end up unable to defend themselves. Proprietary blends cannot be assessed at all, which is itself a finding: the blend is a known unsubstantiable region of the formula, and the claim ladder must treat it that way. Where your form differs from the studied form, note that too; a study on one magnesium salt does not transfer cleanly to another.

Output: a dose comparison table with ratios, citations, and form-match notes.

Done when: every active is classified as at-dose, under-dose, or unassessable.

Claim limits

The compliance envelope, established before any copy exists rather than discovered at review. Three layers, and they are not the same. Regulatory: structure-function claims are permitted with the disclaimer, disease claims are not, and the line between "supports healthy joint function" and "relieves arthritis pain" is the line the whole category is policed on.

Substantiation: what standard of evidence the regulator expects behind each claim, which is where the dose-versus-study output feeds in. Platform: the ad networks apply their own rules, generally stricter than the regulator's, with particular sensitivity to implying knowledge of a user's health condition — before-and-after imagery and second-person symptom language get accounts restricted regardless of legal accuracy.

The output is not a warning. It is a two-column list of permitted and forbidden constructions with the specific language written out, so that downstream work references a document rather than someone's memory of a rule.

Output: permitted and forbidden language lists, plus required disclaimers, per jurisdiction and per platform.

Done when: a copywriter could work from it without asking a follow-up question.

Guard: this is a research artifact, not legal advice. High-risk categories warrant counsel review before spend.

COGS and refills

Landed unit cost, and then the number that a subscription business actually turns on: servings per container divided by servings per day gives days of supply, which gives the natural refill interval. If the bottle runs thirty days and the ingredient takes eight weeks to produce a felt effect, you have a structural churn problem that no amount of creative fixes — customers cancel before the product works. Discovering that here rather than at month three is the whole point of putting this node in research.

Compute gross margin at several candidate price points, including the subscription discount, and note the MOQ and lead time constraints that cap how aggressively you can scale.

Output: landed cost, days of supply, natural refill interval, and margin at candidate prices.

Done when: time-to-effect and refill interval have been explicitly compared and any mismatch is flagged.

The product against the world. This stage produces the go or no-go inputs.

TAM and SAM

Total addressable market for the category, then the serviceable portion — the segment you can actually reach through the channels available to you, at the price you need to charge, under the claim limits you established in stage two. The second number is usually a small fraction of the first and it is the one that matters. Build it upward from population, prevalence, and willingness to pay rather than downward from an industry report headline, because the headline figure includes retail, ingredients you do not sell, and buyers you cannot legally address.

Output: TAM and SAM with the derivation shown, not just the figures.

Done when: the SAM calculation names its channel and price assumptions explicitly.

Offers

What the market is actually being sold, structurally. Price ladders and where each competitor anchors, subscription discount depth, bundle configurations, guarantee terms and length, free-trial and sample mechanics, and the funnel shape each brand uses — advertorial into long-form sales page, direct-to-product, quiz funnel, creator-led.

Read this as a map of what the market has proven it will pay for and in what form. A category where every serious player runs a ninety-day guarantee is telling you something about the time-to-effect problem. A category where nobody discounts subscriptions below fifteen percent is telling you about margin structure.

Output: an offer comparison across the direct set, structural rather than evaluative.

Done when: the dominant offer pattern and the outliers are both identified.

Sophistication

Which of the five stages the market sits at, derived from the offer and creative evidence rather than assessed by feel. Stage one, nobody has made the claim — state it plainly. Stage two, the claim is running — make it bigger or more specific. Stage three, the claims are tired — compete on mechanism. Stage four, mechanisms are competing — elaborate yours or attack theirs. Stage five, everything is exhausted — compete on identity and experience rather than promise.

The evidence test is concrete: count how many competitors are making the same claim, how long it has been running, and how far they have escalated it. Escalation is the tell. Markets do not arrive at stage five gently, they get there because everyone kept raising the promise until nobody believed any of it.

Output: a stage designation with the specific evidence supporting it.

Done when: the stage is defensible by pointing at competitor creative, not by assertion.

Breaking point

What makes someone start looking today rather than continuing not to. Seasonality is the obvious layer and the shallow one. Underneath it sit life events, symptom onset patterns, diagnosis moments, and the failure of whatever they tried before.

Ask the question at its sharpest: not what category of thing prompts a search, but what specific incident made the problem impossible to ignore. What someone said that they cannot unhear. What they saw in a photograph, on a test result, in a shop window, in their child's face. The moment a person decided this cannot go on is both the most emotionally loaded material in the market and the most concrete, and a generic trigger list — "stress", "ageing", "new year" — is what you get when you stop one level too early. Mine these from the same review and forum corpus rather than reasoning them out; they are events people report, not motives you infer.

Output: documented breaking-point moments with circumstances, plus the seasonality curve where one exists.

Done when: each is stated as a specific incident a person could recognise having lived through, and is attributed to a source.

The viability gate

The gate sits here rather than at the end because this is the earliest point at which you have enough to kill the project. TAM, sophistication, offer structure, and margin are all known. Nothing in stages four and five will rescue a market that is at stage five with no mechanism available to you — and running the customer research first, then discovering that, is the expensive version of the same conclusion.

Gate criteria

Size. Is the SAM large enough to support the revenue target at a plausible market share?

Margin. Does the gross margin at a market-acceptable price leave room for a CAC the category actually charges?

Mechanism availability. At the sophistication stage identified, is there a mechanism, ingredient form, or positioning available to you that is not already occupied?

Claim room. Do the claim limits still permit a proposition worth making, given the dose-versus-study results?

Structural churn. Does the refill interval fit the time to effect?

Failing one criterion generally means reposition — a different avatar, a different form factor, a different problem framing, using the adjacent competitor set as the option space. Failing two or more generally means drop. The value of writing this down is that it converts a decision people avoid making into one they have already agreed the terms of.

Only run after the gate clears. This is the most labour-intensive stage and there is no reason to spend it on a market you are not going to enter.

Avatars

Two to four primary avatars, each with its sub-avatars. Build them on three axes and resist collapsing to the first: the situational axis, meaning what is happening in their life that brings them here; the history axis, meaning what they have already tried and how it failed them; and only then the demographic axis. History is the most predictive of the three for supplements and the most frequently omitted — someone who has tried four products and been disappointed needs an entirely different argument from someone shopping the category for the first time, even where they are the same age and income.

Each avatar carries its own pains, desired outcomes, objections, and the identity they hold about themselves in relation to the problem. Two further fields, both of which feed objection handling directly. **Secondary gain:** what the person quietly gets from having the problem and will have to give up to solve it — the excuse it provides, the sympathy it attracts, the effort it exempts them from. A buyer who is unconsciously protecting a secondary gain does not object on price, they object on everything, and the pattern is unreadable unless you have named the gain. **Blame attribution:** who or what they hold responsible — an industry, a doctor, a parent, their own discipline. Whether the blame points outward or inward changes which arguments land and which read as accusation.

Proxy buyers get their own avatar class. A parent buying for a child, an adult child for an ageing parent, one partner for another. This is a distinct buyer, not a variant of the sufferer: the emotional register is more intense — guilt, helplessness, protectiveness — the objections are entirely different, because they concern responsibility for someone else's body rather than their own, and the purchase decision often has to survive a conversation with the person it is for. Supplements attract proxy buying heavily and an avatar set built only around the person with the problem will miss them completely.

Output: avatar profiles with sub-avatars, secondary gain, and blame attribution; proxy buyer avatars where the category has them.

Done when: every avatar can be traced to specific source material, proxy buying has been checked for rather than assumed absent, and the set covers the SAM.

Awareness

Where each avatar sits across the five levels — unaware, problem aware, solution aware, product aware, most aware. Two structural points that the usual treatment obscures. First, it is a distribution, not a value. An avatar spans several levels and what you want is the weighting. Second, it is channel-dependent: the same avatar arrives problem-aware from a cold interest ad and product-aware from a branded search, which means awareness is a property of the intersection rather than of the person.

This determines funnel entry point, and getting it wrong is the most expensive error in the compartment. Product-aware copy served to a problem-aware audience burns spend at scale while looking like a creative problem.

Output: an awareness distribution per avatar per channel.

Done when: each avatar-channel pair has a designated entry level.

Language bank

The surface layer of how people talk. Not summarised sentiment — the literal phrasing, preserved verbatim, organised by function: how they describe the pain, how they describe the outcome they want, what objections they raise, and what proof they demand before believing anything. This is the raw input the copywriter consumes, and it degrades irreversibly the moment it is paraphrased. "I wake up at 3am and can't get back to sleep" is usable; "sleep maintenance issues" is not, and no amount of downstream skill recovers the first from the second.

Keep the pathological detail. Specific times, specific situations, specific failed remedies. Specificity is what makes a reader feel recognised, and it can only come from source material.

Output: a verbatim phrase bank, function-organised, attributed to source.

Done when: each function category holds enough distinct phrasings to write from without reuse.

Identity layer

The layer underneath the language bank, and the one that separates research that produces competent copy from research that produces copy people feel addressed by. Stated desires rarely drive purchase on their own. What drives it is the identity narrative underneath — the label the person carries, the guilt attached to it, the version of themselves they are trying to get back.

Four things to capture, each with a verbatim quote and a source. **Self-applied labels:** what people call themselves in this market — the failure, the tired one, the one who let themselves go. Alongside them, labels others applied that still sting, often from a doctor or a family member. **Guilt and shame:** what they blame themselves for, what they would be mortified for others to know, what they would admit anonymously and never with their name attached. This is the densest material in a proxy-buying category, where the guilt belongs to the purchaser rather than the sufferer. **Loss:** what the problem has cost them — an activity, a role, a relationship, a version of themselves — and what future now feels closed. **Feared identity:** who they are afraid of becoming, usually someone specific they have watched.

Then map the emotional profile per major pain point, on four dimensions rather than one. The primary emotion, named precisely: fear, guilt, shame, grief, anger and helplessness are not interchangeable and produce different copy. The intensity. The duration — whether this is an acute moment, a chronic daily weight, or an episodic flare. And the social dimension: whether the suffering is private, publicly visible, or straining a relationship. That last dimension is the one most often skipped and it matters commercially, because a publicly visible problem carries urgency a private one does not, and a relationship-straining one brings a second person into the purchase decision.

Two cautions specific to this node. It is the easiest place in the entire compartment to write fiction, because plausible identity narratives can be generated for any market without looking at anything — so every line here needs a source or it does not belong. And the material is genuinely intimate. It is collected to describe people accurately, not to manufacture distress; the claim ladder still governs what any of it permits you to say, and shame that a person disclosed in confidence to a forum is not a hook.

Output: labels, guilt and shame narratives, loss narratives, and feared identities, each with a verbatim quote and source link; an emotional profile per major pain point.

Done when: every entry is attributed, and the four emotional dimensions are populated for each major pain point.

Everything above is working material. This stage produces the artifacts that leave the compartment.

Mechanism

Two mechanisms, not one, and the first matters more. The mechanism of the problem explains why nothing they have tried has worked — it reframes their history of failure as a diagnosis rather than a personal deficiency, and it is what earns the right to make a claim in a market where the audience has been disappointed before. The mechanism of the solution then explains why this product addresses that specific cause.

This node requires product truth and sophistication together. At stage three and above, the mechanism is the entire competitive position, and it has to be both true and unoccupied. If no mechanism is available, that is a finding, and it routes back to stage four to look for an avatar for whom a different problem framing applies rather than being papered over with a stronger claim.

Output: problem mechanism and solution mechanism, each in one paragraph a layperson follows.

Done when: both are supported by the ingredient and dose evidence and neither is already running in the competitive set.

LTV and CAC

The subscription arithmetic. An expected churn curve by month rather than a single churn rate, since supplement cancellation clusters heavily at months one and three and an average conceals both cliffs. From the curve, expected retained months, then LTV on contribution margin rather than revenue, then maximum allowable CAC at the payback period you can actually finance.

Payback period is the constraint that decides how the business runs. A ninety-day payback and a first-order payback are different companies with different creative strategies and different scaling ceilings, and the choice belongs here rather than being discovered by running out of cash.

Output: churn curve, LTV, maximum CAC, and payback period, with assumptions listed.

Done when: the maximum CAC is comparable against observed category acquisition costs.

Claim ladder

Every claim you might make, ranked twice: by proof strength, drawing on the dose-versus-study output, and by regulatory risk, drawing on the claim limits. The ladder runs from claims that are both well-substantiated and low-risk at the top, down through claims that are substantiated but risky and claims that are permitted but thin, to claims that are off the table entirely.

The rung matters as much as the claim: the strongest permissible claim usually sits below the strongest true one, and knowing the gap is what lets you choose deliberately rather than drift upward under performance pressure. Nothing downstream gets written that is not on this ladder.

Output: a ranked claim table with substantiation source and risk grade per row.

Done when: every claim carries both grades and the cut line is marked.

Positioning

One fork to settle explicitly, because leaving it implicit is how positioning drifts. Are you selling a **superior product** — it works better, and the argument is mechanism and proof — or an **identity product**, where the argument is who this is for and what buying it says about them? Most of this framework is built for the first, which is the right default for supplements, where efficacy is the claim. But at sophistication stage five the mechanism argument is exhausted and identity is what remains, and the identity layer from stage four is what you build on. Naming the fork forces a choice; leaving it unnamed produces copy that half-argues both and convinces nobody.

Then the big idea and the one-sentence position: who this is for, what it does, and why it is different in a way the buyer cares about. It has to be consistent with the mechanism, sit within the claim ladder, and be defensible against the offer landscape. Include the category entry point — the moment and context in which you want to be the thing that comes to mind — because in a subscription business that is what compounds over time while individual campaigns do not.

Output: big idea, one-sentence position, category entry point.

Done when: the position cannot be truthfully claimed by any competitor in the direct set.

Angle map

The only artifact that leaves the compartment. A matrix of avatar by awareness level by mechanism, where each populated cell is a candidate angle carrying its entry point, the claim rung it is allowed to use, the voice-of-customer language for that avatar, and the objections it must clear.

The reason this is a matrix rather than a document is that it is machine-consumable and coverage-checkable. You can see at a glance which avatar-awareness combinations have no angle and which are overserved, and a creative strategist can pull a cell and brief from it without reading anything else. Everything above it is supporting evidence that stays available but does not need to be read.

Field	Source node
Avatar and sub-avatar	Avatars
Awareness level and channel	Awareness
Mechanism angle	Mechanism
Permitted claim rung	Claim ladder
Language to use	Language bank
Identity narrative engaged	Identity layer
Entry moment	Breaking point
Objections to clear	Review mining, Avatars
Offer and price context	Offers, LTV and CAC
Provenance	Evidenced or inferred

The provenance field is one word per cell and it is the field that keeps the artifact honest. **Evidenced** means every component traces to captured source material. **Inferred** means some part is a hypothesis — a plausible avatar nobody has been observed to be, an identity narrative that fits the category but was not found in it. Inferred angles are legitimate and often the interesting ones; what is not legitimate is losing track of which is which. Downstream, an inferred angle should be tested before it is scaled, and an evidenced angle that fails is telling you something different from an inferred one that fails.

Done when: every avatar-awareness pair is either populated or explicitly marked as not pursued with a reason, and every populated cell carries a provenance mark.

The two feedback loops

New competitor creative reopens stage three. Sophistication is a snapshot and it moves. A competitor launching an escalated claim or a new mechanism changes the stage you are writing against, which invalidates positioning downstream. This wants a standing cadence — a periodic ad library pull — rather than being treated as a one-time task.

No available mechanism sends you back to stage four. When synthesis finds that every viable mechanism is occupied, the correct response is to return to the avatar work and look for a segment for whom a different problem framing applies, not to escalate the claim. Escalation under pressure is precisely how markets reach stage five, and doing it to your own market shortens the runway for everyone including you.

On evidence and inference

Two failure modes account for most bad research documents, and both produce output that reads as authoritative.

The first is unmarked inference. Mined evidence and imagined psychology look identical once they are formatted into the same document with the same headings. A framework that asks "what does this buyer feel guilty about" can be answered by reading four hundred forum posts or by generating a plausible answer from the product description, and nothing in the finished page distinguishes them. This is why the identity layer carries a source requirement per line and why the angle map carries a provenance field. Inference is not the problem — unlabelled inference is. Hypotheses are how you find angles nobody has run; you simply have to know you are holding one.

The second is volume quotas. Fixed targets — forty quotes here, thirty statements there — are a proxy for thoroughness that inverts under pressure. Where a theme genuinely contains twelve good quotes, the honest output is twelve, and a quota of forty makes the honest output non-compliant. Pair a quota with a mandatory source link and the path of least resistance becomes producing plausible citations, which is the specific way research pipelines fail invisibly. Every done-criterion in this document is a saturation test instead: continue until new sources stop producing new themes. That lands in the same place as a quota when the material is rich and tells the truth when it is thin, which is exactly when you need to be told.

A related point on tone. Research into what people are ashamed of, afraid of, or grieving is legitimate and necessary; you cannot write to someone you have not understood. It does not require framing the work as ethically suspect, and

adversarial framing tends to make output more lurid and less accurate rather than more candid. Describe people as they are. What any of it permits you to say is a separate question, and the claim ladder answers it.

Running it as an agent chain

Two properties of this layout matter if the compartment is being automated. First, the stage boundaries are natural checkpoints: each stage produces artifacts that can be validated before the next stage consumes them, which limits how far a bad inference propagates. An agent that concludes stage five is a market at stage two has produced one wrong field, and the gate catches it, rather than producing a coherent and entirely wrong angle map.

Second, the gather-versus-conclude split in stage one is what makes the output reviewable. Because every conclusion cites raw material captured verbatim, a human reviewing the agent's work checks citations rather than re-running the research. That is the difference between an agent that saves time and one that produces work you have to verify from scratch.